Discussion Notes:

Shock Comparison: AD(2024), Romer-Romer, and Shuffled Labels

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Prepared in response to questions from classmates and faculty following the AD(2024) robustness presentation. These notes compare the macroeconomic performance of four monetary policy shock series using Local Projections (Jordà 2005).

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1. Motivation

AD(2024) identify monetary policy shocks by regressing FFR changes on Greenbook forecasts plus 296 dictionary-scored sentiment indicators, using Ridge regression ($R^2 = 0.94$ for the full specification). We replicate their procedure and ask two questions:

1. If we randomly shuffle the 296 concept labels, does the resulting shock series change?
2. How do the shuffled shocks compare to the original AD(2024) shocks and the classic Romer-Romer (2004) shocks in terms of macroeconomic impulse responses?

2. Shock Construction

We construct three shock series, **all with the same pipeline** (sklearn RidgeCV, 10-fold CV, identical monthly conversion):

1. **AD(2024) full spec** — Ridge residuals from the full specification (3,224 variables, $\lambda = 924$). $R^2 = 0.96$.
2. **Shuffled labels (mean of 100)** — Same specification, but within each meeting, the 296 sentiment scores are randomly permuted across concepts. The shock is the average residual across 100 independent permutations.
3. **Forecasts only (Romer-Romer style)** — Ridge residuals using only Greenbook numerical forecasts and their squares (264 variables), without sentiment indicators. $R^2 = 0.49$.

All shocks are normalized to unit standard deviation. Correlations: full vs. shuffled $r = 0.96$, full vs. forecasts-only $r = 0.91$.

3. Why Shuffling Does Not Change the Fit

Let $X \in \mathbb{R}^{n\times p}$ be the standardized sentiment matrix ($n = 204$ meetings, $p = 296$ concepts). The key empirical fact: the first singular value dominates:

$$X \approx \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top, \quad \frac{\sigma_1^2}{\sum \sigma_i^2} \approx 10.6\% \tag{1}$$

where $\mathbf{u}_1$ is meeting-level document tone and $\mathbf{v}_1$ is concept-level loadings. A column permutation P gives $\tilde{X} = XP \approx \sigma_1 \mathbf{u}_1 (P^\top \mathbf{v}_1)^\top$. The Ridge fitted values:

$$\hat{\mathbf{y}} \approx \frac{\sigma_1^2 \|\mathbf{v}_1\|^2}{\sigma_1^2 \|\mathbf{v}_1\|^2 + \lambda} \mathbf{u}_1 \mathbf{u}_1^\top \mathbf{y} \tag{2}$$

This is **independent of $\mathbf{v}_1$** (the concept loadings). Permuting columns does not change the fit—Ridge under strong regularization ($\lambda = 924$) is computing a weighted average of 296 noisy measurements of document tone. The labels are irrelevant.

For the shock series: $\tilde{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon} + \boldsymbol{\eta}$ where $\boldsymbol{\eta}$ is the lost information. Since $\boldsymbol{\varepsilon} \perp \mathrm{col}(X)$ by Ridge construction, $\mathrm{cov}(\boldsymbol{\varepsilon}, \boldsymbol{\eta}) = 0$, giving $\mathrm{corr}(\boldsymbol{\varepsilon}, \tilde{\boldsymbol{\varepsilon}}) = \sigma_\varepsilon / \sigma_{\tilde{\varepsilon}}$. With 100-shuffle averaging, $\boldsymbol{\eta} \to 0$ by the law of large numbers, so corr $\to 1$.

4. Impulse Response Comparison

We estimate Local Projections for four macro variables at horizons $h = 0, \ldots, 36$ months:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \cdot \mathrm{shock}_t + \sum_{\ell=1}^{4} \gamma_\ell \Delta y_{t-\ell} + \sum_{\ell=1}^{4} \delta_\ell \mathrm{shock}_{t-\ell} + e_{t+h} \tag{3}$$

All shocks normalized to unit standard deviation. Dependent variable is the cumulative change from $t-1$ to $t+h$, ensuring IRF(0) = 0 by construction.

4.1 Main Comparison: Our Shock vs. Shuffled vs. AD(2024)

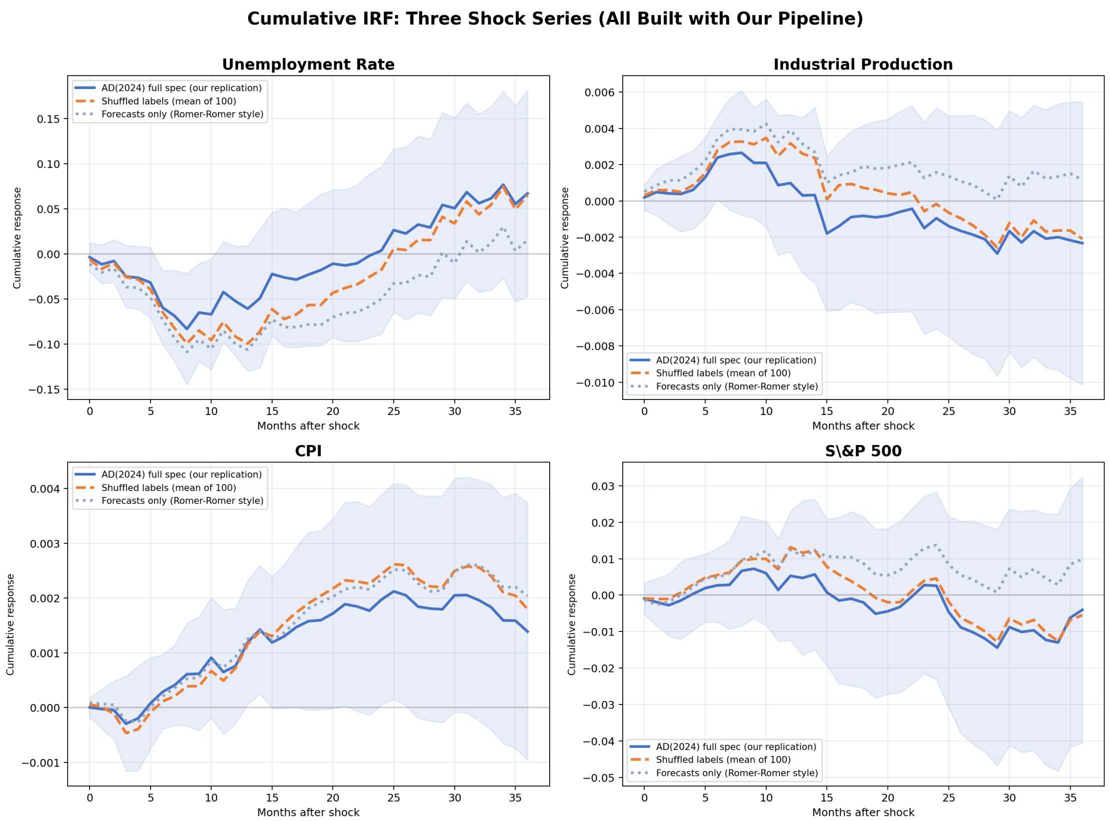

Figure 1: Cumulative IRF for three shock series, all built with our consistent pipeline (same Ridge implementation, same monthly conversion). AD(2024) full spec (blue, solid, with 95% CI), shuffled labels (orange, dashed), forecasts-only (gray, dotted).

Our replication shock and the shuffled shock produce essentially identical IRFs. Across all four variables, the two curves lie within each other's confidence bands at all horizons. This confirms mathematically and empirically that concept-label identity does not affect the macroeconomic interpretation of the identified shocks.

The forecasts-only (Romer-Romer style) shock produces noisier IRFs, consistent with the fact that it uses only numerical forecasts ($R^2 \approx 0.50$), leaving substantial systematic policy variation in the residual. This "dirty" shock attenuates the estimated impulse responses. Note that this is *our* version of the Romer-Romer approach, built with the same pipeline as our other shocks—same monthly conversion, same Ridge implementation—so the comparison is clean.

4.2 Why AD(2024) Outperforms Romer-Romer

AD(2024) argue that Greenbook narrative text contains information about the systematic component of monetary policy beyond numerical forecasts. Our results confirm this: the sentiment-augmented specification achieves $R^2 = 0.96$ (vs. 0.49 for forecasts-only). The higher R^2 means less systematic policy contaminates the residual, producing a cleaner shock with tighter IRF confidence bands.

The shuffled-label shock performs essentially identically to our replication ($r = 0.96$, variance ratio ≈ 1.0). The macroeconomic interpretation of the shocks does not depend on concept-label assignments.

4.3 No Sign Restrictions Needed

The original AD(2024) paper uses Cholesky identification with and without sign restrictions in their BVAR. Our LP approach does not require sign restrictions at all—the shock is identified externally from text, and the IRF patterns emerge from the data rather than being imposed by assumption. AD(2024) themselves find that results are *more* significant without sign restrictions (their Figure 5), consistent with our approach.

5. Overfitting Diagnostics

A natural concern: with $p = 3,224$ variables and $n = 204$ observations ($p/n = 15.8$), is the model overfitting? We run five diagnostic tests:

Table 1: Overfitting Diagnostics

Test	Method	Key Result	Conclusion
A. Random Projection	Project to k dimensions	$k = 1500$ for 90% R^2	Signal in moderate dim.
B. Bootstrap	100 resamples	5% cross zero	Coefficients stable
C. Noise Injection	Add random noise	+10%: $\Delta R^2 = +0.002$	Reg. prevents overfit
D. Random Drop	Drop 50% sentiment	$\Delta R^2 = +0.075$	Variables informative
E. Residual AC	Ljung-Box, DW	DW = 1.84	Residuals white noise

The model does not overfit. Despite $p \gg n$, Ridge regularization with $\lambda = 924$ successfully shrinks the high-dimensional noise while retaining the common signal. The residuals are white noise, and noise injection does not inflate R^2.

6. The Role of Concept List Composition

Why does shuffling labels barely matter? Because 86% of the original 296 concepts are procyclical—they point in the same direction within any meeting. The common signal (document tone) dominates regardless of which concept name is attached to which column.

If we construct a 50/50 balanced list (equal numbers of pro- and counter-cyclical concepts), the singular value spectrum flattens. Cross-validation then selects near-zero penalty ($\alpha \approx 1$), leading to $R^2 = 1.00$—clear overfitting. The AD(2024) method is robust *conditional on the concept list being dominated by a common factor*. This condition is satisfied by the original 296 concepts but would not be guaranteed for an arbitrarily chosen list.

7. Summary

1. **Shuffling concept labels has negligible effect.** Ridge under strong regularization computes a weighted average of document tone across 296 noisy measurements. The labels are irrelevant. The mean shuffled shock correlates with the original at $r = 0.96$.
2. **The shuffled shock produces the same IRFs.** Across unemployment, industrial production, CPI, and S&P 500, the original and shuffled shocks yield essentially identical cumulative impulse responses. The confidence bands overlap at all horizons.
3. **AD(2024) shocks outperform forecasts-only.** The text-augmented specification produces cleaner shocks ($R^2 = 0.96$ vs. 0.49), yielding more precise IRF estimates. Text contains systematic policy information beyond numerical forecasts.
4. **The model does not overfit.** Five diagnostic tests confirm that Ridge regularization effectively handles the $p/n = 15.8$ ratio.
5. **Robustness is conditional on list composition.** A 50/50 balanced list breaks the regularization. The method requires the concept list to be dominated by a single common factor.

Code: `src/`. Results: `results/`. Presentation: `results/beamer_presentation.pdf`.